

PROJECT SUMMARY, INSIGHTS & RECOMMENDATIONS REPORT (PSIR) v1.0

AI vs Human Content Detection System

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1. EXECUTIVE SUMMARY

The AI vs Human Content Detection System was developed to address the growing challenge of identifying AI-generated content across digital platforms. As Large Language Models (LLMs) continue to improve, distinguishing between human-written and machine-generated content has become increasingly difficult for moderation teams.

The primary objective of this project was to design, validate, and deliver a production-ready machine learning solution capable of classifying textual content as either AI Generated or Human Written.

The project successfully progressed through all phases of the machine learning lifecycle including business analysis, exploratory data analysis, feature engineering, model development, validation, leakage investigation, external benchmarking, threshold optimization, serialization, and deployment packaging.

The final approved solution utilizes a hybrid NLP architecture combining Word-Level TF-IDF features, Character-Level TF-IDF features, and Logistic Regression classification.

The solution has been serialized, validated, documented, and packaged for deployment by the client engineering team.

2. BUSINESS PROBLEM OVERVIEW

The client operates a digital content platform that receives large volumes of textual submissions across multiple content categories.

The increasing adoption of generative AI technologies created several challenges:

- Difficulty distinguishing AI-generated content from human-authored content.
- Increased moderation workload.
- Growing risk of content authenticity concerns.

- Reduced scalability of manual review processes.
- Inconsistent moderation decisions across content categories.

The organization required an automated and scalable approach capable of supporting content moderation while preserving platform trust and content integrity.

3. PROJECT OBJECTIVES

The project was initiated with the following objectives:

Primary Objectives

- Detect AI-generated content.
- Protect legitimate human-authored content.
- Reduce manual moderation effort.
- Improve moderation consistency.
- Deliver a production-ready machine learning solution.

Secondary Objectives

- Perform cross-domain validation.
 - Investigate potential data leakage.
 - Support deployment packaging.
 - Enable future scalability.
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4. PROJECT JOURNEY

The project followed a structured machine learning development lifecycle.

Phase 1 – Data Understanding & Validation

The provided dataset was reviewed to understand content structure, available attributes, class distributions, and overall data quality.

Key Outcome:

- Dataset suitability confirmed.
 - No major quality blockers identified.
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Phase 2 – Exploratory Data Analysis

Exploratory analysis was performed to identify trends, relationships, and potential predictive signals.

Key Findings:

- AI-generated content often exhibited more structured and formal language.

- Human-authored content displayed greater stylistic variation.
 - Several metadata attributes appeared highly predictive and required further investigation.
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Phase 3 – Feature Engineering

Both structured features and NLP-based features were evaluated.

Evaluated Features:

- Word Count
- Average Sentence Length
- Domain Information
- Topic Metadata
- TF-IDF Features
- Character-Level Features

Key Outcome:

- NLP-based features demonstrated stronger predictive value than handcrafted structured features.
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Phase 4 – Model Development

Multiple machine learning approaches were explored and evaluated.

Candidate Models Included:

- Logistic Regression
- Random Forest
- XGBoost
- Hybrid Feature Models

Key Outcome:

- Logistic Regression consistently demonstrated strong performance and deployment efficiency when combined with TF-IDF features.
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Phase 5 – Leakage Investigation

During development, several experiments produced near-perfect performance.

This triggered a dedicated leakage investigation process.

Activities Performed:

- Duplicate Content Analysis
- Train/Test Contamination Review

- Metadata Leakage Assessment
- Vocabulary Leakage Assessment

Key Outcome:

- Leakage risks were identified and mitigated.
 - GroupShuffleSplit was implemented to improve validation integrity.
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Phase 6 – External Validation

An independent external validation dataset containing AI-generated and human-authored content was created.

Sources Included:

- Academic Content
- News Content
- Social Content
- GPT Generated Samples
- Gemini Generated Samples

Key Outcome:

- Real-world generalization performance was measured successfully.
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Phase 7 – Threshold Optimization

The classification threshold was optimized to improve operational performance.

Default Threshold:

0.50

Final Threshold:

0.44

Key Outcome:

- Improved AI detection capability while maintaining acceptable precision.
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Phase 8 – Serialization & Deployment Packaging

The final model and supporting artifacts were serialized using Joblib.

Generated Artifacts:

- final_ai_detector.pkl
- decision_threshold.pkl

- model_metadata.pkl

Key Outcome:

- Deployment-ready package delivered.
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5. KEY PROJECT INSIGHTS

Insight 1 – Internal Accuracy Alone Is Not Enough

Several candidate models achieved perfect internal scores.

However, external validation demonstrated that internal performance alone was not a reliable indicator of real-world effectiveness.

Lesson Learned:

External validation is essential when evaluating AI detection systems.

Insight 2 – Leakage Investigation Was Critical

The project identified situations where dataset characteristics could artificially inflate performance.

Implementing GroupShuffleSplit significantly improved confidence in evaluation results.

Lesson Learned:

Validation design is often more important than model complexity.

Insight 3 – Character-Level Features Added Significant Value

Character-level TF-IDF features captured stylistic writing patterns that were not represented by word-level features.

These features improved generalization across unseen content.

Lesson Learned:

Writing style signals can be as valuable as vocabulary signals.

Insight 4 – Metadata Features Introduced Risk

Several metadata fields appeared highly predictive.

Examples:

- source_model
- generation_method
- topic_hint

While useful during experimentation, these features created deployment and leakage concerns.

Lesson Learned:

Features available during training are not always suitable for production use.

Insight 5 – Generalization Matters More Than Accuracy

The final model was selected based on its ability to generalize rather than its internal performance metrics.

Lesson Learned:

A model that performs well on unseen data is more valuable than a model that performs perfectly on known data.

6. FINAL SOLUTION SUMMARY

Approved Production Model

Word + Character TF-IDF Logistic Regression

Configuration

- Word TF-IDF Features
 - Character TF-IDF Features
 - Logistic Regression Classifier
 - Balanced Class Weighting
 - Threshold = 0.44
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7. FINAL PERFORMANCE SUMMARY

Internal Validation

Metric	Score
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Accuracy	100.00%
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Precision	100.00%
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Recall	100.00%
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F1 Score	100.00%
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External Validation

Metric	Score
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Accuracy	82.05%
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Precision	81.82%
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Metric	Score
Recall	91.67%
F1 Score	86.47%

8. BUSINESS IMPACT

The completed solution is expected to provide the following benefits:

Reduced Moderation Effort

Automated classification significantly reduces manual review workload.

Faster Decision Making

Predictions can be generated in near real-time.

Improved Scalability

The solution supports higher content volumes than manual moderation processes.

Consistent Classification

Machine learning-driven decisions improve consistency across content categories.

Deployment Readiness

The solution has been packaged and validated for deployment.

9. RISKS & CONSIDERATIONS

The following risks remain after project completion:

AI Model Evolution

Future AI systems may exhibit different writing characteristics.

Domain Drift

Content topics and styles may change over time.

Data Drift

Language usage patterns may evolve.

Short Content Challenges

Very short content samples may reduce prediction confidence.

10. RECOMMENDATIONS

Short-Term Recommendations (0–3 Months)

- Deploy the current model.
- Monitor prediction quality.

- Review false positive classifications.
 - Capture user feedback.
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Medium-Term Recommendations (3–6 Months)

- Expand external validation datasets.
 - Collect additional AI-generated samples.
 - Perform retraining assessment.
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Long-Term Recommendations (6–12 Months)

- Establish a monitoring framework.
 - Implement a retraining pipeline.
 - Evaluate transformer-based detection models.
 - Expand multilingual support.
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11. FINAL RECOMMENDATION

Based on the completed validation activities, leakage investigation, external benchmarking, threshold optimization, and serialization testing, the AI vs Human Content Detection System is considered suitable for deployment.

The solution demonstrates strong AI detection capability, acceptable generalization performance, and operational readiness for production implementation.

Future enhancements should focus on continuous validation, monitoring, and retraining to maintain effectiveness as generative AI technologies evolve.

12. PROJECT CLOSURE STATEMENT

The AI vs Human Content Detection System successfully completed all planned phases of the machine learning lifecycle and delivered a validated, serialized, and deployment-ready machine learning solution.

The project achieved its primary objective of distinguishing AI-generated content from human-authored content and has been formally prepared for client deployment and operational use.

Project Status: COMPLETED

Deployment Status: READY FOR CLIENT IMPLEMENTATION